Additional Results for

Variational Outlier-Robust Gaussian Process Regression with Generative Modeling

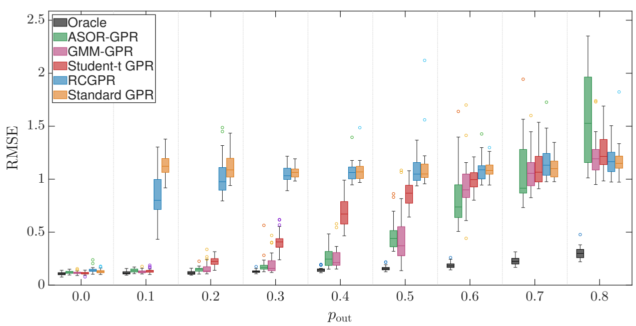
Arslan Majal and Aamir Hussain Chughtai

Additional Prediction RMSE Results

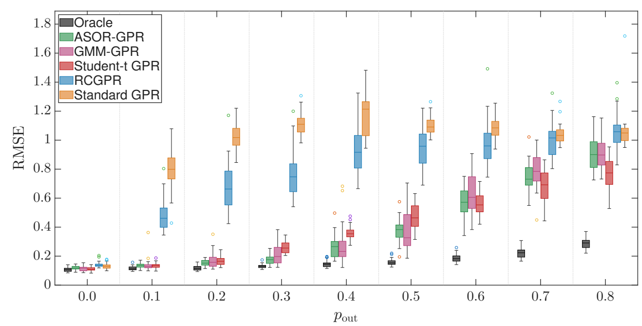
Synthetic-Data Results

This document presents prediction RMSE results under four additional training-target contamination settings. The synthetic-data generation procedure, regression methods, kernel initialization, and evaluation protocol are identical to those described in the main paper.

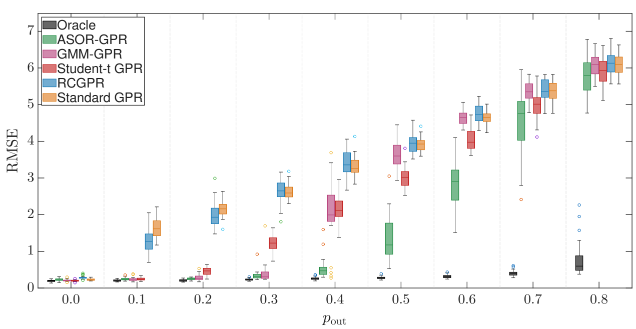
The box plots summarize prediction RMSE over 30 independent Monte Carlo trials. Contamination is injected entry-wise into the training targets only. No non-oracle method receives the true contamination probability p_{out}, the realized contamination indicators, or the generating contamination statistics. Only the Oracle receives this information.



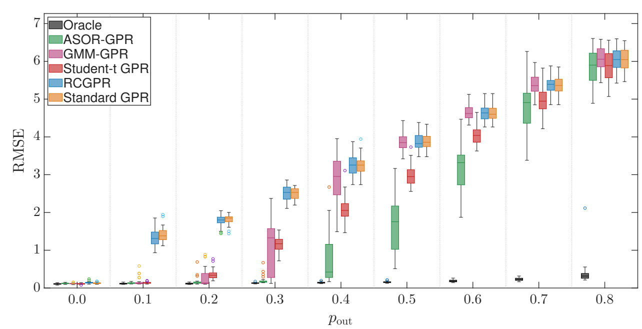
(a) Uniform outliers over $[-10, 10]$.



(b) Gaussian outliers with variance 10.



(c) Uniform outliers over $[0, 15]$.



(d) Uniform outliers over $[-15, 0]$.

Figure 1. Synthetic-data prediction RMSE under different training-target contamination settings. Each box summarizes 30 independent Monte Carlo trials.

Figure 1 compares the methods under Gaussian, symmetric-uniform, and one-sided uniform contamination. ASOR-GPR remains competitive across these settings, indicating that its robustness is not restricted to a particular contamination distribution or direction.

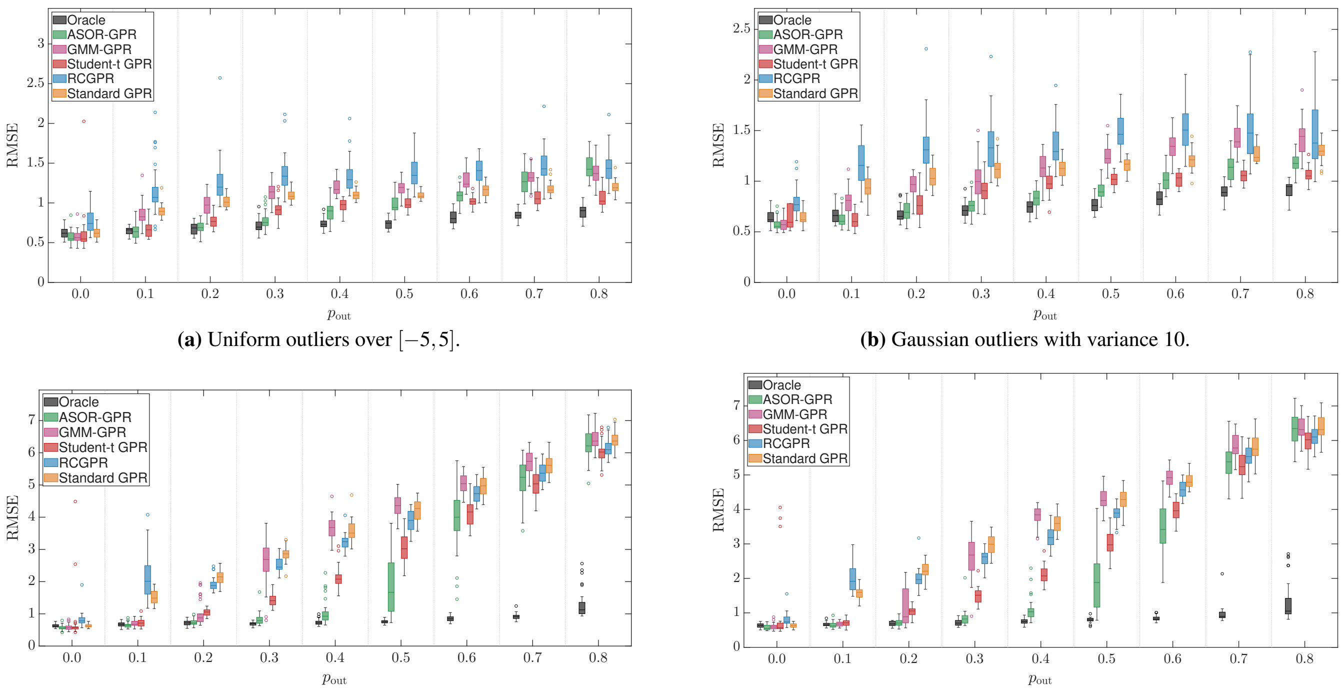
Real-Data Results

We additionally evaluate the compared methods on the Air Quality and Energy Efficiency datasets. For each dataset, the inputs and outputs are standardized using statistics computed from the corresponding training set only. No additional nominal noise is introduced. Artificial contamination is injected entry-wise into the standardized training targets, whereas the standardized test targets remain unmodified and are used to evaluate prediction RMSE.

The box plots summarize prediction RMSE over 30 independent Monte Carlo trials. The Air Quality experiments use three output variables, whereas the Energy Efficiency experiments use the heating-load and cooling-load responses.

As in the synthetic-data experiments, no non-oracle method receives p_{out}, the realized contamination indicators, the generating outlier variance, or the endpoints of the generating uniform distribution. All non-oracle settings are fixed independently of the artificial contamination configuration. Only the Oracle receives the artificial entry-wise contamination indicators and specified component statistics.

Air Quality Dataset



(a) Uniform outliers over $[-5,5]$.

(b) Gaussian outliers with variance 10.

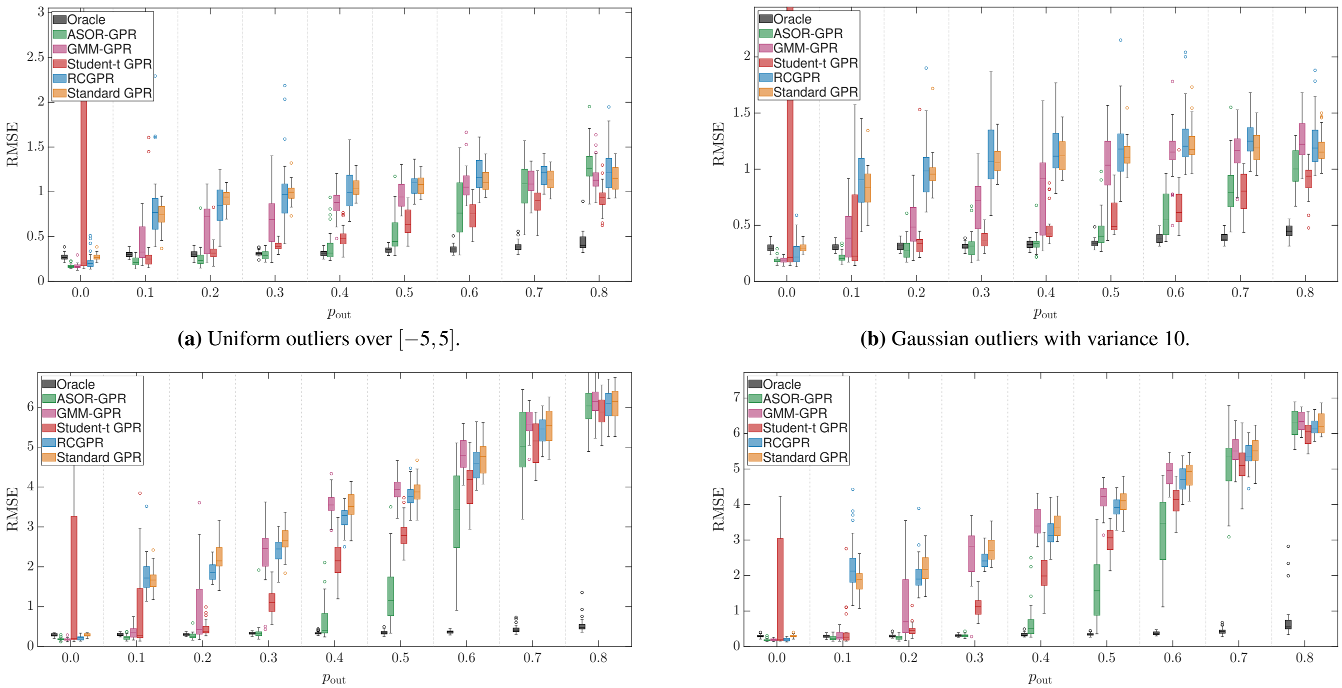
(c) Uniform outliers over $[0,15]$.

(d) Uniform outliers over $[-15,0]$.

Figure 2. Air Quality prediction RMSE under different training-target contamination settings. Each box summarizes 30 independent Monte Carlo trials.

Figure 2 shows the prediction performance under Gaussian, symmetric-uniform, and one-sided uniform contamination. ASOR-GPR remains competitive across the considered settings, indicating that its robustness extends to real data with multiple output variables.

Energy Efficiency Dataset



(a) Uniform outliers over $[-5,5]$.

(b) Gaussian outliers with variance 10.

(c) Uniform outliers over $[0,15]$.

(d) Uniform outliers over $[-15,0]$.

Figure 3. Energy Efficiency prediction RMSE under different training-target contamination settings. Each box summarizes 30 independent Monte Carlo trials.

Figure 3 presents the corresponding results for the heating-load and cooling-load outputs. ASOR-GPR maintains competitive prediction performance across different contamination distributions and output variables.

Model-Fitting Runtime Scaling

We examine the model-fitting runtime of all six regression methods as the training-set size increases. The experiment uses the synthetic-data configuration with four independent scalar-output models, contamination probability $p_{\text{out}} = 0.4$, and zero-mean Gaussian contamination with variance 10. The considered training-set sizes are

$$n_{\text{train}} \in \{200, 400, 600, 800, 1000\}.$$

For every sample size, 30 independent Monte Carlo trials are performed. The experiment is executed serially without a parallel loop, with MATLAB restricted to one computational thread. The recorded runtimes include model fitting only; data generation, prediction, diagnostic collection, plotting, and file saving are excluded.

All methods use independent scalar-output models with the same training inputs, contaminated training targets, and kernel initialization in each Monte Carlo trial. No non-oracle method is provided with the contamination probability, realized contamination indicators, true nominal-noise variance, or generating outlier variance. Only the Oracle receives the exact artificial contamination indicators and corresponding component-noise statistics. ASOR-GPR and GMM-GPR are allowed at most 1000 outer iterations, while all other methods retain their method-specific optimization settings described in the main paper.

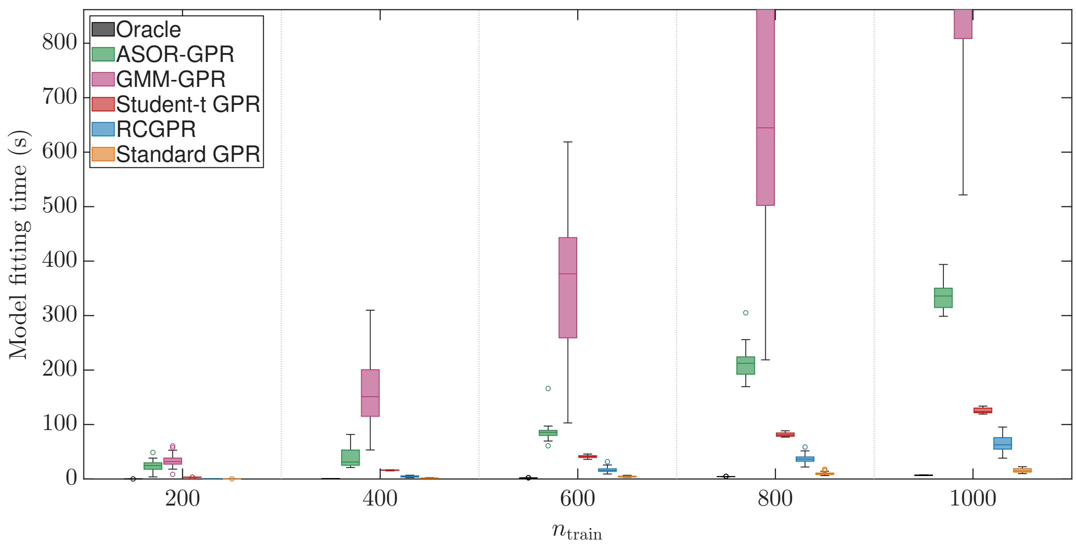


Figure 4. Model-fitting runtime as a function of the training-set size for the Oracle, ASOR-GPR, GMM-GPR, Student-*t* GPR, RCGPR, and standard GPR.

Figure 4 shows that the fitting runtime increases with the training-set size for all methods. The Oracle and standard GPR are the fastest, followed by RCGPR and Student-*t* GPR, whereas ASOR-GPR and GMM-GPR require more time because they repeatedly update latent variables and kernel hyperparameters. Nevertheless, ASOR-GPR is consistently faster and less variable than GMM-GPR. The empirical runtime exponents reflect finite-sample behavior rather than formal asymptotic complexity because runtime also depends on convergence and implementation. The leading dense-matrix operations of all compared GP methods require $\mathscr{O}(n^3)$ time and $\mathscr{O}(n^2)$ memory.